

NODE JS INTRODUCTION

Introduction to Node.js

Node.js is an open-source, cross-platform JavaScript runtime environment that allows developers to run JavaScript code on the server side. It uses the **V8 JavaScript engine**, the same engine that powers Google Chrome, to execute code outside of a browser. Node.js is particularly known for its **non-blocking, event-driven architecture**, making it efficient and suitable for building scalable network applications.

Key Features of Node.js:

1. Asynchronous and Event-Driven:

2. All APIs of Node.js are non-blocking, meaning they run asynchronously. This allows Node.js to handle multiple connections simultaneously.

3. Fast Execution:

Node.js uses the V8 engine, which compiles JavaScript into machine code, leading to faster execution.

4. Single-Threaded but Highly Scalable:

Despite using a single-threaded model, Node.js can handle many connections at once using event loops.

5. Cross-Platform:

Node.js applications can run on various platforms (Windows, Linux, macOS).

6. NPM (Node Package Manager):

Node.js has a large ecosystem of libraries and modules, accessible via npm, which simplifies development.

Setting Up Node.js:

1. Download & Install:

Visit nodejs.org to download the latest stable version.

2. Verify Installation:

Open your terminal/command prompt and type:

```
bash
```

```
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```

```
node -v # Displays the Node.js version
```

```
npm -v # Displays the npm version
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